

Day 08 –CHALLENGE

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TASK 1 — NETWORK IDENTITY

From ipconfig /all on YOUR computer record: IPv4 Address · Subnet Mask · Default Gateway · DNS Server · MAC/Physical Address · DHCP Enabled (Yes/No).

Ipv4 address: 192.168.100.209 , the host portion here is 209 , last usable address is 192.168.199.254

Subnet mask: 255.255.255.0 ,that means the network is c type (prefix – /24)

Default gateway: 192.168.100.1

DNS server: 8.8.8.8 ,this is google's dns server.

MAC/Physical address: C4-13-75-3C-F1-01

DHCP status: enabled — yes

```
PS C:\Windows\System32> ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-AM22QKJ6
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-16
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 789184551
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
NetBIOS over Tcpip. . . . . : Enabled
```

Unknown adapter McAfee VPN:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-74-4B-36-3C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 4:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : C6-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 5:

```
Media State . . . . . : Media disconnected
```

Wireless LAN adapter Wi-Fi 5:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #5
Physical Address. . . . . : CA-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : C4-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e3bb:c8ce:a438:c2e7%5(Preferred)
IPv4 Address. . . . . : 192.168.100.209(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, September 10, 2026 6:48:38 PM
Lease Expires . . . . . : Saturday, September 12, 2026 3:20:24 PM
Default Gateway . . . . . : 192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DHCPv6 IAID . . . . . : 113513333
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
DNS Servers . . . . . : fe80::1%5
                        8.8.8.8
                        8.8.4.4
                        fe80::f912:e97b:33cc:e623%5
                        fe80::1%5
NetBIOS over Tcpip. . . . . : Enabled
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : C4-13-75-3C-F1-02
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

TASK 4 — DNS INVESTIGATION

Run nslookup google.com. State: which DNS server responded, which IP address was returned, and what the purpose of DNS is. Attach a screenshot.

DNS server that responded: google.com

IP address returned: 172.217.23.174 (IPv4)

DNS is used to translate domain names that are easy for humans to understand into ip addresses.

```

PS C:\Windows\system32> nslookup google.com
Server: dev.opt
Address: fe80::1

Non-authoritative answer:
Name: google.com
Addresses: 2a00:1450:4025:1807::65
           2a00:1450:4025:1807::66
           2a00:1450:4025:1807::64
           2a00:1450:4025:1807::71
           172.217.23.174

```

TASK 5 — NETWORK PATH INVESTIGATION

Run `tracert google.com`. State: how many visible hops appeared, what the first hop was, what device commonly represents the first hop, and what traceroute is used for. Attach a screenshot.

10 hops appeared

The first hop is dev.opt : 192.168.100.1

The first hop is usually the router(default gateway)

Traceroute is used to show nodes or hops the packet travelled to reach a specific server

```

PS C:\Windows\system32> tracert google.com

Tracing route to google.com [192.178.194.102]
over a maximum of 30 hops:

  1    3 ms    4 ms    1 ms dev.opt [192.168.100.1]
  2    4 ms    3 ms    5 ms 41.34.4.1
  3    8 ms    3 ms    6 ms 10.45.10.53
  4    5 ms    4 ms    6 ms 10.35.14.2
  5    5 ms    5 ms    4 ms 10.45.28.66
  6    5 ms    11 ms   5 ms TELECOM-EGY.edge1.Marseille3.Level3.net [213.19.202.134]
  7   51 ms   53 ms   50 ms et-0-0-28.edge1.Marseille3.Level3.net [213.19.201.81]
  8    *      *      * Request timed out.
  9   40 ms   41 ms   39 ms 142.250.163.174
 10   49 ms   46 ms   46 ms mu-in-f102.1e100.net [192.178.194.102]

Trace complete.

```